

A FOUR-VERTEX COUNTEREXAMPLE TO CONJECTURE 7.6 OF DEVRIENDT

ABSTRACT. Conjecture 7.6 of Devriendt asserts that a normalized weighted graph with submodular resistance capacity has nonnegative resistance curvature. We give an exact counterexample on the diamond graph $K_4 - e$: the weighting is normalized and its resistance capacity is submodular, but its curvature vector is $(2/5, 2/5, 7/30, -1/30)$. The example is vertex-minimal. It refutes Conjecture 7.6 but does not refute the graph-level Conjecture 7.7.

1. THE COUNTEREXAMPLE

Let $G = (V, E)$ be a connected simple graph with positive conductances c . Write $\Omega = (\omega_{uv})_{u,v \in V}$ for its effective-resistance matrix and $K = \Omega^{-1}$. Following [1], the weighting is *normalized* if

$$\mathbf{1}^\top K \mathbf{1} = 1.$$

For a normalized weighting, the resistance capacity is the set function $\tau : 2^V \rightarrow \mathbb{R}$ defined by

$$\tau_\emptyset = 0, \quad \tau_{\{v\}} = \frac{1}{2}, \quad \tau_U = \frac{1}{2} + \frac{1}{2} \left(\mathbf{1}^\top \Omega[U]^{-1} \mathbf{1} \right)^{-1} \quad (|U| \geq 2),$$

where $\Omega[U]$ is the principal submatrix indexed by U . The resistance curvature is

$$p_v = 1 - \frac{1}{2} \sum_{e \ni v} c_e \omega_e.$$

Conjecture 7.6 of [1] states that, for a normalized weighting, submodularity of τ implies $p \geq 0$.

Theorem 1. *Let $V = \{0, 1, 2, 3\}$ and*

$$E = \{02, 03, 12, 13, 23\},$$

so that G is the diamond graph with nonedge 01 . In the edge order $(02, 03, 12, 13, 23)$, set

$$c = \frac{109}{300}(1, 2, 1, 2, 2).$$

Then c is normalized and τ is submodular, whereas

$$p = \left(\frac{2}{5}, \frac{2}{5}, \frac{7}{30}, -\frac{1}{30} \right).$$

Consequently, Conjecture 7.6 is false.

Proof. The weighted Laplacian is

$$L = \frac{109}{300} \begin{pmatrix} 3 & 0 & -1 & -2 \\ 0 & 3 & -1 & -2 \\ -1 & -1 & 4 & -2 \\ -2 & -2 & -2 & 6 \end{pmatrix}.$$

Using

$$L^\dagger = \left(L + \frac{1}{4} \mathbf{1}\mathbf{1}^\top \right)^{-1} - \frac{1}{4} \mathbf{1}\mathbf{1}^\top$$

and $\omega_{uv} = (e_u - e_v)^\top L^\dagger (e_u - e_v)$ gives

$$\Omega = \frac{1}{109} \begin{pmatrix} 0 & 200 & 140 & 110 \\ 200 & 0 & 140 & 110 \\ 140 & 140 & 0 & 90 \\ 110 & 110 & 90 & 0 \end{pmatrix}.$$

Direct inversion yields

$$K\mathbf{1} = \left(\frac{2}{5}, \frac{2}{5}, \frac{7}{30}, -\frac{1}{30} \right)^\top, \quad \mathbf{1}^\top K\mathbf{1} = 1,$$

so the weighting is normalized. Moreover, in the stated edge order,

$$(c_e \omega_e)_{e \in E} = \frac{1}{15} (7, 11, 7, 11, 9).$$

Substitution in the definition of curvature gives the displayed vector p .

It remains to verify submodularity. For distinct $i, j \in V$ and $S \subseteq V \setminus \{i, j\}$, define the elementary slack

$$\Delta_{ij}(S) = \tau_{S \cup \{i\}} + \tau_{S \cup \{j\}} - \tau_S - \tau_{S \cup \{i, j\}}.$$

A set function is submodular if and only if all such slacks are nonnegative. Put $\Gamma_{ij}(S) = 3924 \Delta_{ij}(S)$. Substitution of Ω into the definition of τ gives the following complete list up to the automorphism $0 \leftrightarrow 1$:

ij	$\{a, b\} = V \setminus \{i, j\}$	$\Gamma_{ij}(\emptyset)$	$\Gamma_{ij}(\{a\})$	$\Gamma_{ij}(\{b\})$	$\Gamma_{ij}(\{a, b\})$
01	$\{2, 3\}$	162	560	165	0
02	$\{1, 3\}$	702	1100	414	249
03	$\{1, 2\}$	972	975	684	124
23	$\{0, 1\}$	1152	864	864	13

The rows for 12 and 13 are obtained from those for 02 and 03 by swapping 0 and 1. Thus the table accounts for all 24 elementary slacks, each of which is nonnegative. Hence τ is submodular, while $p_3 = -1/30 < 0$. $\square$

Proposition 2. *No positively weighted connected simple graph on two or three vertices has negative resistance curvature. Hence four is the smallest possible order of a counterexample.*

Proof. For an edge e , the quantity $c_e\omega_e$ is its inclusion probability in the weighted random spanning tree. It is therefore 1 on a bridge and strictly less than 1 on a nonbridge. The connected simple graphs on two or three vertices are K_2 , P_3 , and K_3 . The first two are trees and have curvature vectors $(1/2, 1/2)$ and $(1/2, 0, 1/2)$, respectively. In K_3 , each vertex is incident to two nonbridges, so the sum of the two incident quantities $c_e\omega_e$ is strictly less than 2 and every curvature coordinate is positive. $\square$

Scope. Conjecture 7.7 of [1] asks whether a graph admits a normalized weighting with submodular resistance capacity exactly when it admits a weighting with nonnegative resistance curvature. It does not require the same weighting in the two existence statements. The diamond graph is resistance positive: unit conductances give curvature

$$\left(\frac{3}{8}, \frac{3}{8}, \frac{1}{8}, \frac{1}{8}\right),$$

and a common rescaling normalizes the weighting without changing curvature. Thus the example refutes precisely the fixed-weight implication in Conjecture 7.6; it does not refute Conjecture 7.7.

REFERENCES

- [1] K. Devriendt, *Graphs with nonnegative resistance curvature*, Ann. Comb. **30** (2026), 415–438. doi:10.1007/s00026-025-00774-x.